Notes: Cremers and Petajisto (RFS, 2009)

How Active Is Your Fund Manager? A New Measure That Predicts Performance

Contents

1 Big picture

- **Question.** How active is an actively managed equity mutual fund, and does the degree and type of active management predict future fund performance?
- **Main innovation.** The paper introduces *Active Share*, a holdings-based measure of how much a fund's portfolio differs from its benchmark index. It is defined as one-half of the sum of absolute differences between fund weights and benchmark weights.
- **Why the paper needs a new measure.** The traditional measure of active management is tracking error: the volatility of fund returns around benchmark returns. Tracking error captures correlated active bets, such as market, sector, style, or industry timing. It does not cleanly capture diversified stock selection, because many individual stock bets can diversify away and produce low tracking error.
- **Two dimensions of active management.** The paper argues that active management is not one-dimensional:
  - **Active Share** emphasizes stock selection: how different the holdings are from the benchmark.
  - **Tracking error** emphasizes factor timing: how volatile the fund's return is relative to the benchmark.
- **Fund types.** Combining the two measures allows the authors to classify funds into pure indexing, closet indexing, diversified stock picking, concentrated stock picking, and factor betting.
- **Empirical setting.** The authors study U.S. all-equity mutual funds from 1980 to 2003. For each fund-date, they compute Active Share relative to nineteen common benchmark indexes from S&P/Barra, Russell, and Wilshire, assigning the benchmark that minimizes Active Share.
- **Main cross-sectional result.** Active Share and tracking error are positively related, but far from equivalent. A fund with a tracking error of 4–6% can have Active Share anywhere from 30% to 100%. This means the two measures identify different forms of activity.
- **Main time-series result.** The mutual fund industry became less active over time. The share of fund assets with Active Share below 60% rose from 1.5% in 1980 to 44.8% in 2003. The share with Active Share above 80% fell from 42.8% to 23.3%.
- **Main performance result.** Active Share predicts benchmark-adjusted fund performance. In 1990–2003, the highest Active Share funds outperform their benchmarks by about 1.13–1.15% per year net of fees, while the lowest Active Share nonindex funds underperform by about 1.42–1.83% per year. The high-minus-low Active Share spread is about 2.55–2.98% per year.
- **What does not work.** Tracking error by itself does not predict higher fund performance. Funds that look active mainly through high tracking error often do not add value.
- **Interpretation.** Some managers appear to have stock-picking skill, and this skill is visible among high Active Share funds. Closet indexers charge active-management fees while delivering portfolios close to benchmark indexes, so they underperform net of fees.
- **Economic takeaway.** Investors should not ask only whether a fund is labeled active. They should ask whether it is *truly different* from its benchmark, because only genuine deviations from the benchmark create the possibility of benchmark outperformance.

2 Data and measurement (key objects)

2.1 Data sources

- **Mutual fund holdings.** Holdings come from the CDA/Spectrum mutual fund holdings database maintained by Thomson Financial. The database reports mutual fund holdings, mostly quarterly, beginning in 1980.
- **Mutual fund returns.** Monthly fund returns come from CRSP. These are net returns after expenses, fees, and brokerage commissions, but before front-end or back-end loads.
- **Daily fund returns.** Daily fund returns come mainly from Standard & Poor's Micropal/Worths data for live funds, supplemented with CRSP daily returns after 2001 and Wall Street Web data for dead funds before 2001. Daily returns are needed to estimate tracking error more accurately.
- **Benchmark holdings.** Benchmark holdings are collected for nineteen common indexes from S&P/Barra, Russell, and Wilshire. This is a key data contribution because the paper does not force every fund to use the same benchmark.
- **Stock-level match.** Fund and benchmark stock holdings are matched to CRSP stock-return data.

2.2 Benchmark universe

The benchmark set includes:

- S&P 500, S&P 500/Barra Growth, S&P 500/Barra Value, S&P MidCap 400, and S&P SmallCap 600;
- Russell 1000, Russell 2000, Russell 3000, Russell Midcap, and the value and growth components of each;
- Wilshire 5000 and Wilshire 4500.

Interpretation. This benchmark universe matters because Active Share is meaningful only relative to the correct investment universe. A small-cap fund should not be measured mechanically against the S&P 500 if its natural benchmark is the Russell 2000 or S&P SmallCap 600.

2.3 Sample selection

The sample is constructed to focus on U.S. all-equity mutual funds:

- exclude bond funds, balanced funds, asset-allocation funds, international funds, sector funds, precious-metal funds, and other non-broad-equity funds;
- require reported holdings to align with benchmark holdings dates;
- require enough daily returns to compute tracking error: at least 100 trading days in the six months before the holdings report date;
- require equity holdings greater than $10 million;

- handle multiple share classes by aggregating total net assets and value-weighting expense ratios, loads, turnover, and stock-share variables.

The final sample contains 2,647 funds from 1980 to 2003 and 48,354 fund-report-date observations.

2.4 Active Share

For fund f and benchmark b, Active Share is

$$\text{ActiveShare}_{f,b,t} = \frac{1}{2} \sum_{i=1}^{N} \left| w_{f,i,t}^{fund} - w_{b,i,t}^{index} \right|, \tag{1}$$

where $w_{f,i,t}^{fund}$ is the weight of stock i in fund f, and $w_{b,i,t}^{index}$ is the weight of stock i in benchmark index b.

Interpretation.

- Active Share equals the fraction of the fund that is different from the benchmark.
- A pure index fund has Active Share close to 0%.
- A fund with zero overlap with its benchmark has Active Share close to 100%.
- The division by 2 avoids double-counting the implicit active long side and active short side of the same benchmark deviation.

2.5 Portfolio-decomposition interpretation

Any fund portfolio can be written as

$$w_i^{fund} = w_i^{index} + a_i, \qquad \sum_i a_i = 0, \tag{2}$$

where a_i is the active weight in stock i.

Then

$$\text{ActiveShare} = \frac{1}{2} \sum_i |a_i| = \sum_{a_i > 0} a_i = -\sum_{a_i < 0} a_i. \tag{3}$$

Interpretation. A mutual fund can be viewed as holding 100% of the benchmark plus a zero-net-investment active long-short portfolio. Active Share is the size of that active long-short portfolio.

2.6 Tracking error

The standard tracking-error measure is

$$\text{TrackingError}_f = \text{Stdev}\left(R_{f,t}^{fund} - R_{b,t}^{index} \right). \tag{4}$$

The paper uses a modified version based on a regression of excess fund returns on excess benchmark returns:

$$R_{f,t}^{fund} - R_{f,t}^{riskfree} = \alpha_f + \beta_f \left(R_{b,t}^{index} - R_{f,t}^{riskfree} \right) + \varepsilon_{f,t}, \tag{5}$$

$$\text{TrackingError}_f = \text{Stdev}(\varepsilon_{f,t}). \tag{6}$$

Interpretation. This version does not mechanically punish a fund for having a persistent benchmark beta different from one or for holding a stable cash position. It focuses on residual volatility around the benchmark relation.

2.7 Benchmark assignment

For each fund-date, the assigned benchmark is the index with the lowest Active Share:

$$b^*_{f,t} = \arg\min_{b \in \mathcal{B}} \text{ActiveShare}_{f,b,t}, \tag{7}$$

where $\mathcal{B}$ is the set of nineteen candidate indexes.

Interpretation.

- The chosen benchmark is the index whose holdings overlap most with the fund.
- If the assignment is "wrong" relative to a fund's stated benchmark, it is wrong only in the sense that the fund's actual holdings look closer to another index.
- This method avoids relying on self-declared benchmarks, which are unavailable for much of the sample and may be strategically chosen.

2.8 Net returns, gross returns, and performance metrics

- **Net returns** are actual investor returns after fees and transaction costs.
- **Gross returns** are hypothetical returns on disclosed portfolio holdings, excluding fees and transaction costs.
- **Benchmark-adjusted returns** subtract the fund's assigned benchmark return.
- **Four-factor alphas** adjust benchmark-adjusted returns or excess returns using the Carhart market, size, value, and momentum factors.
- **Characteristic Selectivity** follows Daniel et al. (1997) and measures stock-selection performance from holdings.

Interpretation. Gross returns ask whether the manager chose a portfolio that beat the benchmark before costs. Net returns ask whether investors actually benefited after costs.

3 Framework and empirical specifications (all equations + interpretation)

3.1 Two-dimensional classification of active management

The paper's conceptual framework is:

	Low tracking error	**High tracking error**
High Active Share	Diversified stock picker	Concentrated stock picker
Low Active Share	Closet indexer	Factor bettor
Very low Active Share and very low tracking error	Pure index fund	

Interpretation.

- A diversified stock picker can have high Active Share but low tracking error because stock-specific bets diversify.
- A factor bettor can have high tracking error but lower Active Share because broad sector or style tilts create correlated return deviations.
- A closet indexer has low Active Share and low tracking error while still being sold as active management.

3.2 Why Active Share differs from tracking error

Let active weights be $a_i = w_i^{fund} - w_i^{index}$. Ignoring beta adjustment for intuition, benchmark-relative returns are

$$R_t^{fund} - R_t^{index} = \sum_i a_i R_{i,t}. \tag{8}$$

Tracking error depends on the covariance matrix:

$$\text{TrackingError}^2 = a'\Sigma a. \tag{9}$$

Active Share depends only on the absolute active weights:

$$\text{ActiveShare} = \frac{1}{2}\sum_i |a_i|. \tag{10}$$

Interpretation. Tracking error weights correlated active bets more heavily. Active Share weights all active positions equally, even if their risks diversify. This is why Active Share is closer to a stock-selection measure and tracking error is closer to a factor-timing measure.

3.3 Industry-level Active Share

To isolate broad industry bets, the authors also describe an industry-level version of Active Share:

$$\text{ActiveShare}_{f,b,t}^{industry} = \frac{1}{2}\sum_{j=1}^{10} \left| w_{f,j,t}^{fund} - w_{b,j,t}^{index} \right|, \tag{11}$$

where j indexes ten industry portfolios.

Interpretation. If a fund's stock-level Active Share is high but industry-level Active Share is low, the fund is making within-industry stock picks rather than large sector bets.

3.4 Determinants of Active Share

The fund-year regression is conceptually:

$$\begin{aligned}\text{ActiveShare}_{f,t} = \gamma_t + \beta_1 \text{TrackingError}_{f,t} + \beta_2 \text{Turnover}_{f,t} + \beta_3 \text{Expenses}_{f,t} \\ + \beta_4 \log(\text{TNA}_{f,t}) + \beta_5 \log(\text{TNA}_{f,t})^2 + \theta' X_{f,t} + u_{f,t}.\end{aligned} \tag{12}$$

where $X_{f,t}$ includes the number of stocks, fund age, manager tenure, fund inflows, prior benchmark-adjusted returns, and prior benchmark returns.

Interpretation. This regression asks whether Active Share is just a relabeling of size, turnover, fees, age, flows, or tracking error. The answer is no: tracking error is related to Active Share, but much variation remains unexplained.

3.5 Benchmark-adjusted performance

A simple benchmark-adjusted return is

$$r_{f,t}^{BA} = R_{f,t}^{fund} - R_{b_{f,t}^*,t}^{bench}. \tag{13}$$

The four-factor benchmark-adjusted alpha is estimated from

$$r_{f,t}^{BA} = \alpha_f^{BA} + \beta_M \text{MKT}_t + \beta_S \text{SMB}_t + \beta_H \text{HML}_t + \beta_U \text{UMD}_t + \epsilon_{f,t}. \tag{14}$$

Interpretation. Benchmark adjustment asks whether the fund beats its closest low-cost passive alternative. The four-factor correction checks whether the benchmark-adjusted returns still reflect exposure to market, size, value, or momentum factors.

3.6 Two-dimensional fund sorts

Each month in the 1990–2003 performance sample:

1. exclude index funds;
2. sort active funds into Active Share quintiles;
3. within those groups, sort into tracking-error quintiles;
4. compute equal-weighted fund portfolio returns;
5. estimate benchmark-adjusted returns and four-factor benchmark-adjusted alphas.

Interpretation. The double sort asks whether stock-selection activity or factor-timing activity is more closely associated with future performance.

3.7 Predictive performance regression

The multivariate predictive regression is conceptually:

$$\begin{aligned}\text{Perf}_{f,t+1} = \gamma_t + \beta \text{ActiveShare}_{f,t} + \delta \text{ActiveShare}_{f,t} \cdot \mathbf{1}\{\text{Small}_{f,t}\} \\ + \lambda \text{TrackingError}_{f,t} + \theta' Z_{f,t} + \varepsilon_{f,t+1}.\end{aligned} \tag{15}$$

where $\text{Perf}_{f,t+1}$ is next year's performance measure and $Z_{f,t}$ includes fees, turnover, size, number of stocks, age, manager tenure, flows, and prior returns.

Interpretation. This specification asks whether Active Share predicts future performance after controlling for other fund characteristics and whether the relation is stronger for smaller funds.

4 Identification and empirical strategy (what makes the estimates credible)

4.1 What identifies the main result?

The paper's performance claim is not based on a single time-series event. It is based on cross-sectional variation in how much funds differ from their benchmarks.

- Funds with higher Active Share are compared to funds with lower Active Share.
- Benchmark-adjusted returns are measured relative to each fund's assigned benchmark.
- Performance is examined using both gross returns and net returns.
- The analysis controls for tracking error, fund size, fees, turnover, and other fund characteristics.

Interpretation. The core comparison is: among active funds, do funds that deviate more from their benchmarks perform better than funds that stay close to their benchmarks?

4.2 Why benchmark assignment is central

A fund's Active Share can be artificially high if measured against the wrong index. The authors reduce this concern by choosing the benchmark that minimizes Active Share among nineteen realistic alternatives.

Strength. The benchmark is determined from holdings, not from marketing labels or fund self-reports.

Limitation. A fund may have a legitimate blended benchmark, but the main analysis assigns one benchmark. The authors note this as a possible extension.

4.3 Why benchmark-adjusted returns are emphasized

The paper argues that benchmark-adjusted returns are the natural object because a fund investor can usually buy the benchmark cheaply through index funds, ETFs, futures, or similar products. The active manager's job is to beat that benchmark.

Important nuance. Standard Carhart alphas of fund excess returns do not show the same Active Share relation because passive benchmark indexes themselves have nonzero Carhart alphas during the sample. The authors interpret this as evidence of factor-model misspecification rather than benchmark skill.

4.4 Why gross and net returns are both useful

- **Gross returns** reveal whether managers pick stocks that beat benchmarks before costs.
- **Net returns** reveal whether the skill survives fees and trading costs.

Interpretation. High Active Share funds show positive gross performance, and the best groups also beat benchmarks net of costs. Closet indexers do not display skill and underperform net of costs.

4.5 Important limitations

- Active Share is not a sufficient condition for performance. A fund can be very different from its benchmark and still make bad bets.
- Holdings are periodic, so the measure observes disclosed portfolios rather than continuous trading.
- The main sample ends in 2003; the paper documents the historical emergence of closet indexing through that date, not the whole modern ETF era.
- The predictive relation is strongest for benchmark-adjusted performance. Standard factor-model alphas can behave differently because of benchmark-factor misspecification.
- The analysis is about broad U.S. all-equity funds, not sector funds, international funds, bond funds, or hedge funds.

5 Main results and interpretation

5.1 Active Share and tracking error are related but distinct

In the 2002 cross-section, there is wide dispersion in both dimensions. The most important empirical fact is that neither measure subsumes the other.

- A tracking error of 4–6% can correspond to Active Share from 30% to 100%.
- Active Share of 70–80% can correspond to tracking error from 2% to above 14%.
- This validates the two-dimensional framework empirically.

Economic meaning. Some funds are active because they pick many individual stocks while diversifying away sector risk. Other funds are active because they take broad factor or industry bets.

5.2 Small funds are more active, but only weakly at first

High Active Share funds tend to be smaller, but the relation is nonlinear and not as strong as a simple capacity story would suggest.

- For large-cap funds in 2002, marginal Active Share remains around 70% from roughly \$10 million to \$1 billion in assets.

- Above $1 billion, Active Share falls more quickly.
- This suggests that funds do not simply index all marginal inflows. They often scale up existing active positions.

Interpretation. Size matters, but it is not enough to explain Active Share. Fund strategy and manager behavior matter too.

5.3 Fees are not strongly tied to true activity

Index funds have low fees, but among nonindex funds, expense ratios vary surprisingly little with Active Share.

- Equal-weighted expense ratio across all funds in 2002: 1.24%.
- Lowest Active Share and lowest tracking-error group: 0.47%.
- Highest Active Share funds: about 1.42%.
- Closet-indexing groups charge fees much closer to active funds than to index funds.

Interpretation. Closet indexers are especially expensive because investors pay active fees for benchmark-like portfolios.

5.4 Turnover is not the same as active management

Turnover is weakly correlated with Active Share. Many nonindex fund groups have similar turnover, while pure index funds have much lower turnover.

Interpretation. Trading frequently is not the same as holding a portfolio that differs meaningfully from the benchmark. Turnover measures activity in trading, while Active Share measures activity in positions.

5.5 Closet indexing increased over time

The paper documents a major time trend toward lower Active Share.

- Assets in funds with Active Share below 60% rose from 1.5% in 1980 to 44.8% in 2003.
- Assets in funds with Active Share above 80% fell from 42.8% in 1980 to 23.3% in 2003.
- Index funds grew rapidly after 1990 and reached 15.3% of assets in 2003.
- Nonindex funds with Active Share below 60% were rare before 1987 but reached roughly 30% of assets around 2000–2001.

Interpretation. The 1990s saw both a rise in explicit indexing and a rise in implicit indexing inside nominally active funds.

5.6 Aggregate active positions partially cancel out

When the authors aggregate the holdings of S&P 500-benchmarked active funds into one large portfolio, aggregate Active Share is much lower than the average fund-level Active Share.

- In the 1980s, value-weighted fund-level Active Share is about 75–80%, while aggregate Active Share is about 45%.
- In the most recent years of the sample, value-weighted fund-level Active Share is about 55–60%, while aggregate Active Share is about 30%.

Interpretation. Roughly half of active fund positions offset one another inside the mutual fund sector. Some funds' active long positions are other funds' active underweights.

5.7 Active Share predicts performance

In the 1990–2003 nonindex-fund sample, Active Share is strongly related to benchmark-adjusted performance.

- Highest Active Share funds earn 1.13% per year benchmark-adjusted net return; lowest Active Share funds earn −1.42%.
- The high-minus-low spread is 2.55% per year.
- With four-factor benchmark-adjusted alphas, highest Active Share funds earn 1.15% and lowest Active Share funds earn −1.83%, a spread of 2.98%.
- Gross returns show that the highest Active Share funds beat benchmarks before costs by 2.40% per year and by 1.51% using four-factor benchmark-adjusted alphas.

Interpretation. High Active Share is associated with genuine stock-picking skill; low Active Share active funds tend to deliver negative net value after fees.

5.8 Tracking error does not predict performance

Tracking error does not generate the same positive performance relation. In many specifications, high tracking error is unrelated or negatively related to performance.

Interpretation. Being volatile relative to a benchmark is not the same as being skilled. Factor timing is harder to monetize than stock picking, or the broad factors are priced too efficiently for active managers to exploit.

5.9 The Active Share relation is stronger for smaller funds

The performance spread between high and low Active Share funds is strongest in the smaller size quintiles.

- Four-factor benchmark-adjusted alpha spread between high and low Active Share: 3.78% in the smallest fund-size quintile.
- Corresponding spreads in size quintiles 2 and 3: 3.20% and 3.05%.

- Spread in the largest size quintile: 1.01%, no longer statistically significant.

Interpretation. Smaller active funds appear more able to convert active stock selection into abnormal performance. Large funds face capacity constraints or are more likely to hug benchmarks.

5.10 Performance persistence is strongest among high Active Share funds

High Active Share funds with strong prior-year benchmark-adjusted gross returns continue to perform well.

- Within the highest Active Share quintile, the four-factor alpha spread between prior-year winners and losers is 4.48% per year.
- Prior-year winners in the highest Active Share quintile earn a 3.50% annualized four-factor benchmark-adjusted net alpha.
- Their benchmark-adjusted net return is 5.10% per year.
- Among below-median-sized funds, the top group earns 6.49% benchmark-adjusted net return and 4.84% four-factor benchmark-adjusted alpha.

Interpretation. Past performance is more informative when the fund is truly active. For closet indexers, past outperformance is less likely to reflect persistent skill.

6 Tables and figures

6.1 Figure 1: Different types of active and passive management

Read: Figure 1 is the conceptual map. Active Share is on one axis, tracking error on the other.

- High Active Share and low tracking error: diversified stock picks.
- High Active Share and high tracking error: concentrated stock picks.
- Low Active Share and high tracking error: factor bets.
- Low Active Share and low tracking error: closet indexing.
- Near zero on both dimensions: pure indexing.

Why it matters. The figure explains why one measure of active management is not enough.

6.2 Table 1: All-equity mutual funds in 2002 sorted by Active Share and tracking error

Read: The table reports the number of funds and median assets in each Active Share/tracking-error cell.

- There are 1,678 all-equity funds in the 2002 cross-section.
- Funds are spread across both dimensions, not concentrated on a single line.

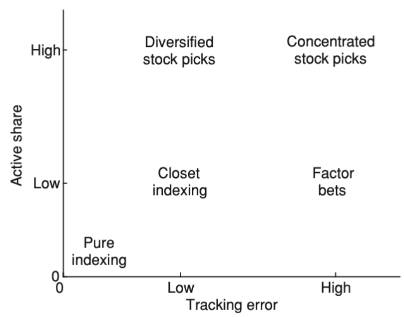

Figure 1
Different types of active and passive management
Active Share represents the fraction of portfolio holdings that differ from the benchmark index, thus emphasizing stock selection. Tracking error is the volatility of fund return in excess of the benchmark, so it emphasizes bets on systematic risk.

Table 1
All-equity mutual funds in the United States in 2002, sorted by the two dimensions of active management

	Tracking error (% per year)								
Active Share (%)	0–2	2–4	4–6	6–8	8–10	10–12	12–14	>14	All
				Panel A: Number of mutual funds					
90–100			66	125	77	41	22	26	358
80–90		17	100	120	54	24	10	10	336
70–80		26	124	83	27	5	7	10	281
60–70		75	115	41	12		1		247
50–60	3	102	55	15	3				179
40–50	9	66	20						98
30–40	15	27	3						47
20–30	11	4							14
10–20	8								10
0–10	104	4							109
All	150	323	482	388	174	73	41	48	1678
				Panel B: Median net asset value ($M)					
90–100			174	127	97	95	93	51	115
80–90		264	263	177	109	128	206	43	183
70–80		264	163	149	185	237	96	26	156
60–70		379	256	208	180				259
50–60		262	256	257					251
40–50	151	304	281						275
30–40	535	269							267
20–30	200								196
10–20	58								58
0–10	480								480
All	395	303	220	162	110	110	92	48	190

Active Share is defined as the percentage of a fund's portfolio holdings that differ from the fund's benchmark index. It is computed based on Spectrum mutual fund holdings data and index composition data for nineteen common benchmark indexes from S&P, Russell, and Wilshire. Tracking error is defined as the annualized standard deviation of the error term when the excess return on a fund is regressed on the excess return on its benchmark index. It is computed based on daily fund returns and daily index returns over a six-month period before the corresponding portfolio holdings are reported. To include only all-equity funds, every fund classified by CRSP as balanced or asset allocation has been removed from the sample. Also sector funds have been eliminated. In panel A, if a cell has less than five observations (fund-dates), it is shown as empty. In panel B, a statistic must be based on at least five funds to be reported.

- Median fund size tends to be lower among high Active Share funds and higher among low Active Share funds.
- Funds with Active Share below 20% are treated as pure index funds.

Why it matters. Table 1 establishes that Active Share and tracking error are empirically distinct, not just two versions of the same statistic.

6.3 Figure 2: Average Active Share and marginal Active Share by fund size

Read: The figure plots Active Share against fund size for large-cap nonindex funds in 2002.

- Average Active Share is broadly stable for small and medium funds.
- It declines more quickly above roughly $1 billion in assets.
- Marginal Active Share remains high for much of the size distribution, contrary to a simple story in which new inflows are indexed.

Why it matters. The figure challenges the strongest version of the capacity argument. Size matters, but it does not automatically imply that all new money becomes passive.

6.4 Table 2: Expense ratios and turnover

Read: The table reports expense ratios and turnover across the two-dimensional active-management grid.

- Average equal-weighted expense ratio is 1.24%.
- Pure index funds are cheap: 0.47% equal-weighted expense ratio in the lowest Active Share/low tracking-error cell.

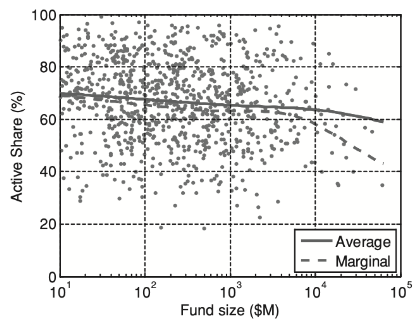

Figure 2
Average Active Share and the Active Share of a marginal dollar
The sample includes U.S. large-cap all-equity mutual funds in 2002. Fund size is total net assets expressed in millions of dollars. We exclude all index funds and funds with less than $10 million in stock holdings. The average Active Share is estimated from a nonparametric kernel regression with a Gaussian kernel and bandwidth equal to 0.5.

Table 2
Expense ratios and annual portfolio turnover for all-equity mutual funds in 2002

	Tracking error (% per year)								
Active Share (%)	0–2	2–4	4–6	6–8	8–10	10–12	12–14	>14	All
Panel A: Equal-weighted total expense ratio (%)									
90–100			1.33	1.37	1.51	1.47	1.49	1.50	1.42
80–90		1.30	1.30	1.43	1.44	1.43	1.37	2.11	1.41
70–80		1.19	1.29	1.37	1.33	1.40	1.85	1.34	1.33
60–70		1.10	1.24	1.35	1.37				1.23
50–60		1.04	1.21	1.43					1.14
40–50	1.12	1.08	1.07						1.08
30–40	1.03	1.06							1.08
20–30	0.92								0.88
10–20	0.71								0.75
0–10	0.47								0.47
All	0.62	1.08	1.27	1.39	1.44	1.45	1.54	1.59	1.24
Panel B: Equal-weighted turnover (%)									
90–100			71.2	101.7	107.8	118.2	140.0	198.5	108.8
80–90		93.5	101.9	133.5	124.5	134.1	210.2	147.5	123.5
70–80		69.3	91.7	98.6	133.8	80.7	74.2	123.9	96.1
60–70		69.0	93.9	107.5	108.0				89.2
50–60		65.5	92.0	87.4					76.8
40–50	57.1	69.7	61.6						67.3
30–40	72.9	117.4							97.8
20–30	141.7								148.9
10–20	60.0								66.1
0–10	18.1								18.4
All	38.2	73.9	89.9	111.1	116.5	119.1	145.3	170.0	94.8

Funds are sorted by the two dimensions of active management. The measures of active management are computed as before. Turnover is defined by CRSP as the maximum of annual stock purchases and annual stock sales, divided by the fund's total net assets. To include only all-equity funds, every fund classified by CRSP as balanced or asset allocation has been removed from the sample. Also sector funds have been eliminated. To be reported in the table, a statistic must be based on at least five funds.

- Highest Active Share funds charge around 1.42%.
- Nonindex low Active Share funds charge fees much closer to active funds than to index funds.
- Turnover is much lower for index funds, but not strongly monotonic in Active Share among nonindex funds.

Why it matters. The table shows why closet indexing can be bad for investors: active-level fees without active-level benchmark deviation.

6.5 Table 3: Determinants of Active Share

Read: Table 3 runs fund-year regressions of Active Share on tracking error and other fund characteristics.

- Tracking error is the strongest predictor; with year dummies, the coefficient is about 1.81.
- A 5% increase in annualized tracking error implies roughly a 9% increase in Active Share.
- Expense ratios and size have statistically significant but economically limited effects.
- Turnover is not an important determinant.
- Inflows do not explain Active Share once current size and other variables are included.
- The broadest specification has R^2 around 32%, so much of Active Share remains unexplained by standard fund characteristics.

Why it matters. Active Share is a genuinely new measure. It cannot be fully replaced by tracking error, size, fees, or turnover.

6.6 Figure 3: Active Share over time

Read: Figure 3 shows the asset share of U.S. all-equity mutual funds in Active Share bins from 1980 to 2003.

Table 3
Determinants of Active Share for all-equity mutual funds in 1992–2003

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Tracking error	1.4015	1.8111	1.7002	1.5965	1.5210	1.4439	
	(19.16)	(18.15)	(17.40)	(16.09)	(12.81)	(12.17)	
Turnover				−0.0016		−0.0021	
				(0.65)		(0.66)	
Expenses				4.4359	4.6230	4.6267	7.7859
				(6.33)	(5.28)	(5.33)	(9.72)
lg(TNA)			0.0554	0.0601	0.0451	0.0614	0.0389
			(2.96)	(3.16)	(2.02)	(2.87)	(1.62)
$(\lg(\text{TNA}))^2$			−0.0177	−0.0171	−0.0150	−0.0177	−0.0166
			(4.85)	(4.58)	(3.56)	(4.36)	(3.65)
Number of stocks						−0.0001	
						(2.04)	
Fund age						−0.0005	−0.0003
						(2.26)	(1.06)
Manager tenure						0.0036	0.0041
						(6.72)	(7.00)
Inflow, $t-1$ to t						0.0052	0.0045
						(1.30)	(1.04)
Inflow, $t-3$ to $t-1$						0.0010	0.0019
						(0.94)	(1.53)
Return over index, $t-1$ to t					0.1068	0.0996	0.1189
					(8.12)	(7.45)	(8.21)
Return over index, $t-3$ to $t-1$					0.1103	0.1089	0.1478
					(9.39)	(9.17)	(13.00)
Index return, $t-1$ to t						0.0655	0.0756
						(5.28)	(6.02)
Index return, $t-3$ to $t-1$					−0.0619	−0.0570	−0.0469
					(7.87)	(6.93)	(5.19)
Year dummies	No	Yes	Yes	Yes	Yes	Yes	Yes
N	11,726	11,726	11,726	11,554	8,417	8,320	8,374
R^2	0.1316	0.2373	0.2642	0.2781	0.2984	0.3235	0.2037

The dependent variable is Active Share for each fund-year observation. All the variables are computed as before. Turnover and expense ratio are annualized values. Fund age and fund manager tenure are measured in years. Fund inflows and returns are all cumulative percentages. Index return represents the benchmark assigned to each fund, and return over the index represents a fund's net return (after all expenses) in excess of its benchmark index. Index funds are excluded from the sample. Since the expense ratio and manager tenure are missing before 1992, we limit all specifications to the same time period. Year fixed-effects are included in all specifications. The t-statistics (in parentheses) are based on standard errors clustered by fund.

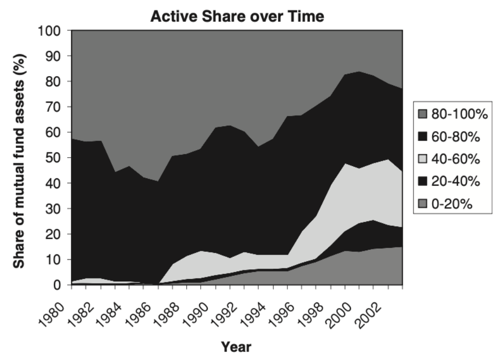

Figure 3
The share of US all-equity mutual fund assets in each Active Share category in 1980–2003

- Active Share below 60% rises from 1.5% of assets in 1980 to 44.8% in 2003.
- Active Share above 80% falls from 42.8% of assets to 23.3%.
- Index funds grow rapidly after 1990, reaching 15.3% of assets in 2003.
- Closet indexing becomes economically meaningful in the 1990s.

Why it matters. The figure documents an industry-wide shift away from truly active management.

6.7 Figure 4: Aggregate-level and fund-level Active Share

Read: Figure 4 compares fund-level Active Share with aggregate Active Share for S&P 500-benchmarked active funds.

- Value-weighted fund-level Active Share is around 75–80% in the 1980s, but aggregate Active Share is around 45%.
- Later in the sample, value-weighted fund-level Active Share is about 55–60%, while aggregate Active Share is about 30%.

Why it matters. Mutual funds trade against each other. The average investor in the active mutual fund sector receives less aggregate active exposure than the average fund-level Active Share suggests.

6.8 Table 4: Net performance by Active Share and tracking error

Read: Table 4 sorts active funds by Active Share and tracking error and reports net benchmark-adjusted performance.

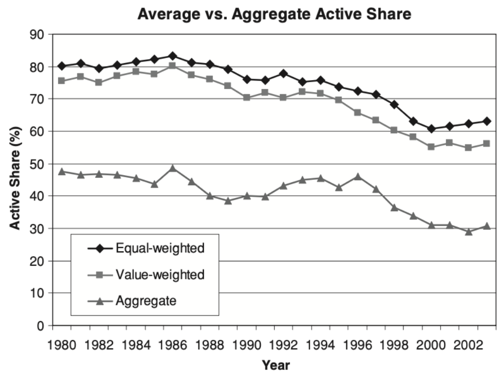

Figure 4
Aggregate-level and fund-level Active Share
We only include active funds that have the S&P 500 as their benchmark index. Each year we compute the equal-weighted and value-weighted (by fund size) Active Share across the funds. We also aggregate the funds' portfolios into one aggregate portfolio and compute its Active Share. Index funds are excluded from the sample.

Table 4
Net equal-weighted alphas for all-equity mutual funds in 1990–2003

Active Share quintile	Tracking error quintile						
	Low	2	3	4	High	All	High-Low
	Panel A: Benchmark-adjusted return						
High	0.09	0.39	1.34	2.76	1.05	1.13	0.97
	(0.09)	(0.41)	(1.52)	(2.86)	(0.62)	(1.60)	(0.44)
4	−0.43	−0.15	0.56	0.50	0.76	0.25	1.20
	(−0.61)	(−0.19)	(0.64)	(0.42)	(0.36)	(0.31)	(0.48)
3	−1.42	−0.98	−0.25	−0.49	−0.60	−0.75	0.82
	(−2.06)	(−1.34)	(−0.29)	(−0.45)	(−0.35)	(−0.95)	(0.43)
2	−1.89	−1.14	−1.13	−1.01	−1.66	−1.37	0.23
	(−3.20)	(−1.55)	(−1.53)	(−1.08)	(−1.22)	(−1.99)	(0.16)
Low	−1.35	−1.32	−1.28	−1.51	−1.63	−1.42	−0.28
	(−4.95)	(−3.68)	(−2.77)	(−2.76)	(−2.13)	(−3.53)	(−0.39)
All	−1.00	−0.64	−0.15	0.05	−0.42	−0.43	0.58
	(−1.92)	(−1.24)	(−0.24)	(0.06)	(−0.30)	(−0.76)	(0.36)
High-Low	1.44	1.71	2.62	4.26	2.68	2.55	
	(1.50)	(1.71)	(2.97)	(4.36)	(1.80)	(3.47)	
	Panel B: Four-factor alpha of benchmark-adjusted return						
High	1.44	0.79	0.48	2.72	0.29	1.15	−1.15
	(1.79)	(1.02)	(0.68)	(3.17)	(0.22)	(1.86)	(−0.74)
4	−0.11	−0.91	−0.88	−1.52	−1.64	−1.02	−1.53
	(−0.22)	(−1.17)	(−1.23)	(−1.63)	(−1.33)	(−1.63)	(−1.08)
3	−1.05	−1.41	−1.58	−2.25	−2.86	−1.83	−1.81
	(−1.97)	(−2.15)	(−2.34)	(−2.23)	(−2.51)	(−2.84)	(−1.59)
2	−1.46	−1.47	−1.82	−2.67	−3.43	−2.18	−1.97
	(−3.31)	(−2.29)	(−2.99)	(−3.31)	(−3.61)	(−4.00)	(−2.17)
Low	−1.29	−1.36	−1.66	−2.26	−2.57	−1.83	−1.28
	(−4.80)	(−4.80)	(−4.33)	(−4.43)	(−3.73)	(−5.01)	(−2.13)
All	−0.50	−0.87	−1.09	−1.20	−2.05	−1.14	−1.55
	(−1.45)	(−2.13)	(−2.58)	(−1.81)	(−2.28)	(−2.53)	(−1.68)
High-Low	2.73	2.16	2.13	4.99	2.86	2.98	
	(3.33)	(2.52)	(2.61)	(5.60)	(2.26)	(4.51)	

Funds are sorted by the two dimensions of active management. The measures of active management are computed as before. Net fund returns are the returns to a fund investor after fees and transaction costs. Index funds are excluded from the sample. The table shows annualized returns, followed by t-statistics (in parentheses) based on White's standard errors.

- Average active fund benchmark-adjusted net return: −0.43% per year.
- Highest Active Share quintile: 1.13% per year.
- Lowest Active Share quintile: −1.42% per year.
- High-minus-low Active Share spread: 2.55% per year.
- With four-factor benchmark-adjusted alphas, the spread rises to 2.98%.

Why it matters. Active Share predicts investor-level performance after fees and trading costs.

6.9 Table 5: Gross performance by Active Share and tracking error

Read: Table 5 repeats the same exercise using gross holdings-based returns.

- Highest Active Share funds beat benchmarks by 2.40% per year before costs.
- Lowest Active Share funds roughly match benchmarks before costs.
- High-minus-low gross benchmark-adjusted spread: 2.29%.
- Four-factor benchmark-adjusted gross alpha for the highest Active Share funds: 1.51%.

Why it matters. The gross results show that high Active Share funds have stock-selection skill before costs; the net results show that some of the skill survives costs.

6.10 Table 6: Fund size and Active Share

Read: Table 6 sorts first by fund size and then by Active Share.

- Active Share predicts performance especially among smaller funds.

Table 5
Gross equal-weighted alphas for all-equity mutual funds in 1990–2003

Active Share quintile	Tracking error quintile: Low	2	3	4	High	All	High-Low
	Panel A: Benchmark-adjusted return						
High	1.34	1.56	3.01	3.34	2.72	2.40	1.38
	(1.61)	(1.67)	(3.30)	(2.70)	(1.29)	(2.80)	(0.60)
4	1.02	1.32	1.35	1.39	1.58	1.33	0.56
	(1.56)	(1.59)	(1.28)	(0.97)	(0.64)	(1.28)	(0.20)
3	0.09	0.78	0.97	1.19	1.06	0.81	0.97
	(0.16)	(1.03)	(1.10)	(1.00)	(0.54)	(0.94)	(0.46)
2	−0.24	0.13	0.68	0.59	0.02	0.24	0.26
	(−0.54)	(0.23)	(0.93)	(0.59)	(0.01)	(0.34)	(0.17)
Low	0.00	0.37	0.06	0.22	−0.08	0.11	−0.08
	(−0.02)	(1.24)	(0.14)	(0.40)	(−0.10)	(0.29)	(−0.10)
All	0.44	0.83	1.22	1.35	1.05	0.98	0.61
	(1.04)	(1.56)	(1.82)	(1.42)	(0.63)	(1.41)	(0.34)
High-Low	1.35	1.19	2.95	3.13	2.81	2.29	
	(1.68)	(1.27)	(3.47)	(2.79)	(1.70)	(3.05)	
	Panel B: Four-factor alpha of benchmark-adjusted return						
High	1.39	0.86	1.38	2.50	1.37	1.51	−0.02
	(1.80)	(1.00)	(1.76)	(2.54)	(1.02)	(2.23)	(−0.02)
4	0.42	−0.23	−0.92	−1.28	−1.08	−0.63	−1.50
	(0.76)	(−0.27)	(−1.07)	(−1.12)	(−0.78)	(−0.81)	(−1.01)
3	−0.33	−0.38	−0.88	−1.24	−1.58	−0.89	−1.26
	(−0.55)	(−0.52)	(−1.15)	(−1.19)	(−1.40)	(−1.28)	(−1.08)
2	−0.59	−0.83	−0.60	−1.58	−2.21	−1.17	−1.63
	(−1.42)	(−1.59)	(−0.98)	(−1.93)	(−2.32)	(−2.14)	(−1.73)
Low	−0.30	−0.13	−0.67	−0.72	−1.31	−0.63	−1.01
	(−1.18)	(−0.47)	(−1.62)	(−1.41)	(−1.98)	(−1.73)	(−1.69)
All	0.12	−0.14	−0.34	−0.46	−0.97	−0.36	−1.09
	(0.32)	(−0.28)	(−0.63)	(−0.62)	(−1.06)	(−0.69)	(−1.18)
High-Low	1.69	0.99	2.05	3.22	2.68	2.13	
	(2.15)	(1.12)	(2.57)	(3.41)	(2.21)	(3.29)	

Funds are sorted by the two dimensions of active management. The measures of active management are computed as before. Gross fund returns are the returns on a fund's portfolio and do not include any fees or transaction costs. Index funds are excluded from the sample. The table shows annualized returns, followed by t-statistics (in parentheses) based on White's standard errors.

Table 6
Net equal-weighted alphas for all-equity mutual funds in 1990–2003

Active Share quintile	Fund size quintile: Low	2	3	4	High	All	High-Low
	Four-factor alpha of benchmark-adjusted return						
High	1.71	1.39	1.15	0.19	−0.67	0.75	−2.39
	(1.97)	(1.58)	(1.61)	(0.25)	(−1.01)	(1.26)	(−2.60)
4	0.87	−0.24	−0.02	−1.55	−1.90	−0.57	−2.78
	(1.09)	(−0.28)	(−0.02)	(−1.72)	(−2.61)	(−0.82)	(−3.74)
3	−1.47	−1.60	−2.11	−2.58	−1.54	−1.86	−0.07
	(−2.21)	(−2.13)	(−2.97)	(−3.71)	(−2.36)	(−3.09)	(−0.12)
2	−1.95	−2.52	−2.79	−1.56	−2.16	−2.20	−0.21
	(−3.24)	(−4.27)	(−4.56)	(−2.04)	(−3.88)	(−4.02)	(−0.43)
Low	−2.06	−1.81	−1.90	−1.64	−1.69	−1.82	0.38
	(−3.97)	(−4.04)	(−4.61)	(−3.89)	(−5.30)	(−4.81)	(1.03)
All	−0.59	−0.96	−1.14	−1.43	−1.60	−1.14	−1.01
	(−1.34)	(−1.93)	(−2.41)	(−2.58)	(−3.41)	(−2.53)	(−3.05)
High-Low	3.78	3.20	3.05	1.83	1.01	2.57	
	(3.74)	(3.22)	(3.75)	(2.51)	(1.43)	(3.87)	

Funds are sorted by fund size and Active Share (sequentially and in that order). Active Share is computed as before. Net fund returns are the returns to a fund investor after fees and transaction costs. Index funds are excluded from the sample. The table shows annualized returns, followed by t-statistics (in parentheses) based on White's standard errors.

- Four-factor benchmark-adjusted alpha spreads between high and low Active Share are 3.78%, 3.20%, and 3.05% in the first three size quintiles.
- The spread falls to 1.83% in size quintile 4 and 1.01% in the largest size quintile.
- Fund size itself is negatively related to performance.

Why it matters. The best signal is not Active Share alone, but Active Share combined with capacity: high Active Share works better among smaller funds.

6.11 Table 7: Active Share and performance persistence

Read: Table 7 sorts by Active Share and prior one-year benchmark-adjusted gross return.

- Within the highest Active Share quintile, prior-year winners continue to outperform.
- Highest Active Share/prior-winner group: 3.50% four-factor benchmark-adjusted net alpha.
- The prior-winner minus prior-loser spread within high Active Share funds is 4.48%.
- Persistence is much weaker among low Active Share funds.

Why it matters. Past performance is more credible when the fund is truly active enough for past performance to reflect manager choices.

6.12 Table 8: Fund alphas and benchmark alphas

Read: Table 8 compares standard Carhart alphas of fund excess returns with Carhart alphas of the funds' benchmark indexes.

- Standard excess-return Carhart alphas show no significant Active Share pattern.

Table 7
Net equal-weighted alphas for all-equity mutual funds in 1990−2003

Active Share quintile	Prior one-year return quintile: Low	2	3	4	High	All	High-Low
	Four-factor alpha of benchmark-adjusted return						
High	−0.98	0.39	0.89	1.04	3.50	0.96	4.48
	(−1.01)	(0.46)	(1.25)	(1.27)	(3.29)	(1.56)	(3.06)
4	−2.28	−1.83	−1.90	−0.44	0.46	−1.19	2.74
	(−2.26)	(−2.14)	(−2.84)	(−0.59)	(0.41)	(−1.90)	(1.72)
3	−2.48	−2.65	−2.08	−1.78	−0.99	−2.00	1.49
	(−2.39)	(−3.24)	(−2.67)	(−2.50)	(−1.04)	(−3.10)	(1.00)
2	−2.55	−2.16	−2.41	−1.77	−1.80	−2.14	0.75
	(−2.47)	(−3.21)	(−4.60)	(−3.14)	(−2.30)	(−3.90)	(0.59)
Low	−2.05	−2.26	−1.78	−1.52	−1.58	−1.84	0.47
	(−3.41)	(−5.85)	(−4.97)	(−3.42)	(−3.14)	(−5.01)	(0.70)
All	−2.07	−1.72	−1.47	−0.90	−0.11	−1.26	1.96
	(−2.69)	(−3.08)	(−3.30)	(−1.99)	(−0.15)	(−2.76)	(1.74)
High-Low	1.07	2.65	2.67	2.56	5.08	2.80	
	(1.15)	(3.14)	(3.45)	(2.67)	(4.91)	(4.33)	

Funds are sorted by Active Share and prior one-year return (sequentially and in that order). The prior return on a fund is measured as its benchmark-adjusted gross return over the previous twelve months. Only funds with at least nine months of such returns are included. Active Share is computed as before. Net fund returns are the returns to a fund investor after fees and transaction costs. Index funds are excluded from the sample. The table shows annualized returns, followed by t-statistics (in parentheses) based on White's standard errors.

Table 8
Equal-weighted Carhart alphas of mutual funds and their benchmark indexes in 1990−2003

Active Share quintile	Tracking error quintile: Low	2	3	4	High	All	High-Low
	Panel A: Four-factor alpha of excess return on funds						
High	0.08	0.13	−0.93	0.72	−1.29	−0.25	−1.37
	(0.06)	(0.10)	(−0.69)	(0.55)	(−0.73)	(−0.21)	(−0.80)
4	−1.40	−0.33	−1.37	−2.34	−1.13	−1.31	0.26
	(−1.41)	(−0.28)	(−1.37)	(−1.83)	(−0.76)	(−1.32)	(0.15)
3	−1.28	−0.24	−0.90	−1.77	−1.17	−1.08	0.11
	(−1.37)	(−0.27)	(−1.09)	(−1.64)	(−0.97)	(−1.35)	(0.07)
2	−1.29	−0.35	−0.62	−1.22	−1.65	−1.03	−0.36
	(−2.11)	(−0.49)	(−1.08)	(−1.64)	(−1.77)	(−2.05)	(−0.32)
Low	−0.65	−0.16	−0.30	−0.41	−0.88	−0.47	−0.23
	(−1.76)	(−0.41)	(−0.60)	(−0.78)	(−1.38)	(−1.24)	(−0.35)
All	−0.91	−0.19	−0.82	−1.00	−1.22	−0.83	−0.32
	(−1.25)	(−0.25)	(−1.18)	(−1.17)	(−1.18)	(−1.19)	(−0.27)
High-Low	0.73	0.29	−0.63	1.13	−0.41	0.23	
	(0.67)	(0.24)	(−0.51)	(0.91)	(−0.25)	(0.21)	
	Panel B: Four-factor alpha of excess return on benchmark indexes						
High	−1.36	−0.66	−1.40	−2.00	−1.58	−1.40	−0.22
	(−1.07)	(−0.54)	(−1.16)	(−2.01)	(−1.64)	(−1.36)	(−0.20)
4	−1.28	0.57	−0.49	−0.82	0.51	−0.29	1.79
	(−1.26)	(0.67)	(−0.61)	(−0.99)	(0.67)	(−0.43)	(1.53)
3	−0.23	1.17	0.68	0.48	1.69	0.76	1.92
	(−0.31)	(1.99)	(1.16)	(0.77)	(2.33)	(1.70)	(1.61)
2	0.17	1.12	1.20	1.45	1.79	1.15	1.61
	(0.33)	(2.28)	(2.19)	(2.54)	(2.62)	(2.83)	(1.72)
Low	0.64	1.20	1.36	1.86	1.69	1.35	1.05
	(1.70)	(2.91)	(2.75)	(2.93)	(2.57)	(3.05)	(1.69)
All	−0.41	0.68	0.27	0.20	0.82	0.31	1.23
	(−0.60)	(1.19)	(0.51)	(0.36)	(1.45)	(0.66)	(1.43)
High-Low	−2.00	−1.86	−2.76	−3.85	−3.27	−2.75	
	(−1.72)	(−1.55)	(−2.22)	(−3.36)	(−2.69)	(−2.53)	

Funds are sorted by the two dimensions of active management. The measures of active management are computed as before. Panel A shows the four-factor alphas of net fund returns in excess of the risk-free rate. Index funds are excluded from the sample. Panel B shows the four-factor alphas of returns on the official benchmark indexes of the funds in panel A. The table shows annualized returns, followed by t-statistics (in parentheses) based on White's standard errors.

- Benchmark indexes of high Active Share funds have negative Carhart alphas.
- Benchmark indexes of low Active Share funds have positive Carhart alphas.
- The benchmark-alpha spread is about 2.75% per year.

Why it matters. The result warns against mechanically interpreting Carhart alphas as skill when passive benchmark indexes themselves can have nonzero alphas.

6.13 Table 9: Multivariate predictive regressions

Read: Table 9 predicts next-year performance from Active Share and other fund characteristics.

- Active Share coefficient for benchmark-adjusted alphas: 0.0722.
- A 30% increase in Active Share predicts a 2.17% increase in next-year benchmark-adjusted alpha.
- When interacted with below-median size, the effect is stronger for smaller funds.
- Tracking error is marginally negative.
- Expenses and size are important negative predictors.

Why it matters. Active Share survives controls; it is not only a univariate sorting artifact.

6.14 Table 10: Comparison with other active-management measures

Read: Table 10 compares Active Share with tracking error, industry-level Active Share, Industry Concentration Index, stock concentration, and turnover.

- Active Share is the measure that most consistently predicts benchmark-adjusted performance.

Table 9
Predictive regression for fund performance in 1992–2003

	Benchmark-adjusted alphas		Excess return (over risk-free rate) alphas		Characteristic selectivity	
	(1)	(2)	(3)	(4)	(5)	(6)
Active Share	0.0722	0.0666	0.0496	0.0450	0.0200	0.0154
	(2.53)	(2.42)	(1.89)	(1.75)	(1.66)	(1.31)
Active Share		0.0139		0.0133		0.0136
(below median size)		(2.36)		(2.55)		(5.07)
Tracking error	−0.1454	−0.1459	−0.1006	−0.1022	−0.0567	−0.0578
	(1.76)	(1.75)	(1.71)	(1.72)	(0.66)	(0.68)
Turnover	−0.0041	−0.0041	−0.0033	−0.0033	0.0003	0.0003
	(0.84)	(0.84)	(0.78)	(0.78)	(0.10)	(0.10)
Expenses	−1.3117	−1.3024	−1.3447	−1.3348	−0.2797	−0.2694
	(6.00)	(5.99)	(5.10)	(5.06)	(0.75)	(0.73)
$\log_{10}$(TNA)	−0.0187	−0.0013	−0.0127	0.0040	−0.0019	0.0155
	(3.06)	(0.14)	(2.22)	(0.43)	(0.31)	(2.39)
$(\log_{10}(\text{TNA}))^2$	0.0025	0.0003	0.0015	−0.0005	0.0003	−0.0019
	(2.12)	(0.22)	(1.46)	(0.40)	(0.23)	(1.60)
Number of stocks/100	0.0043	0.0043	0.0027	0.0028	0.0014	0.0015
	(3.81)	(3.82)	(2.90)	(2.97)	(3.59)	(3.70)
Fund age/100	−0.0213	−0.0214	−0.0129	−0.0130	−0.0088	−0.0089
	(3.45)	(3.44)	(3.67)	(3.59)	(1.80)	(1.80)
Manager tenure/100	0.0245	0.0250	0.0199	0.0201	0.0452	0.0453
	(0.85)	(0.87)	(0.75)	(0.77)	(2.50)	(2.50)
Inflow, $t-1$ to t	0.0025	0.0024	−0.0018	−0.0019	0.0033	0.0033
	(0.77)	(0.74)	(0.53)	(0.56)	(0.88)	(0.85)
Inflow, $t-3$ to $t-1$	0.0010	0.0010	0.0011	0.0012	0.0012	0.0012
	(1.00)	(1.05)	(1.05)	(1.09)	(1.39)	(1.44)
Return over index, $t-1$ to t	0.0745	0.0739	0.0757	0.0753	0.0572	0.0566
	(1.36)	(1.35)	(1.51)	(1.50)	(1.52)	(1.51)
Return over index, $t-3$ to $t-1$	−0.0353	−0.0350	−0.0378	−0.0374	−0.0749	−0.0746
	(1.72)	(1.68)	(1.59)	(1.56)	(3.23)	(3.18)
Index return, $t-1$ to t	0.0369	0.0371	0.0010	0.0011	0.0325	0.0327
	(1.40)	(1.42)	(0.12)	(0.13)	(3.31)	(3.24)
Index return, $t-3$ to $t-1$	−0.0233	−0.0228	−0.0563	−0.0560	−0.0559	−0.0556
	(0.65)	(0.64)	(2.80)	(2.82)	(5.45)	(5.42)
N	8,232	8,232	8,232	8,232	6,615	6,615
R^2	0.0376	0.0395	0.1768	0.1778	0.168	0.1694

The dependent variable in columns 1–4 is based on the cumulative net return (after all expenses) over calendar year t, while the independent variables are measured at the end of year $t-1$. Alphas are computed with respect to the four-factor model. All explanatory variables are computed as before in Table 3. Index funds are excluded from the sample. Since the expense ratio and manager tenure are missing before 1992, we limit all specifications to the same time period. All specifications include year dummies, and columns 3–6 also include benchmark dummies. The t-statistics (in parentheses) are based on standard errors clustered by year.

Table 10
Predicting returns with different measures of active management in 1992–2003

	Benchmark-adjusted alphas								Excess return (over risk-free rate) alphas	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Active Share	0.0560						0.0751	0.0612	0.0067	0.0172
	(2.70)						(2.39)	(2.11)	(0.32)	(0.73)
Active Share							0.0115	0.0141	0.0123	0.0130
(below median size)							(2.26)	(2.40)	(3.07)	(2.52)
Tracking error		−0.0604					−0.2017	−0.1463	−0.1870	−0.1181
		(0.62)					(1.76)	(1.77)	(2.06)	(1.88)
Industry-level Active Share			0.0396				−0.0283	0.0464	0.0303	0.0845
			(1.82)				(0.56)	(0.93)	(0.60)	(1.83)
Industry Concentration Index				0.0816			0.1321	−0.0755	0.0606	−0.1243
				(2.74)			(1.84)	(1.13)	(0.84)	(2.61)
Stock concentration index					0.0388		−0.3070	−0.0503	0.0763	0.1434
					(0.18)		(1.18)	(0.18)	(0.31)	(0.58)
Turnover						−0.0019	−0.0021	−0.0039	−0.0007	−0.0029
						(0.35)	(0.43)	(0.78)	(0.14)	(0.69)
Control variables	No	No	No	No	No	No	No	Yes	No	Yes
N	11,480	11,480	11,480	11,480	11,481	11,351	11,351	8,232	11,351	8,232
R^2	0.0098	0.0004	0.0032	0.0031	0.0002	0.0007	0.0184	0.0397	0.1381	0.1791

The dependent variable is the benchmark-adjusted cumulative net return (after all expenses) over calendar year t, while the independent variables are measured at the end of year t-1. Active Share and tracking error are computed as before. Industry-level Active Share is computed similarly to Active Share, except that it replaces individual stocks with ten industry portfolios. The Industry Concentration Index is computed as in Kacperczyk, Sialm, and Zheng (2005), except that the benchmark index is selected by following the same methodology as elsewhere in this article. The stock concentration index is computed just like the Industry Concentration Index, except that it uses individual stocks rather than industry portfolios. Turnover is an annualized value. The control variables include all the remaining variables in Table 9. Index funds are excluded from the sample. Since the expense ratio and manager tenure are missing before 1992, we limit all specifications to the same time period. All specifications include year dummies, and columns 9–10 also include benchmark dummies. The t-statistics (in parentheses) are based on standard errors clustered by year.

- Tracking error does not predict higher performance and often has a negative sign.
- Industry concentration measures can matter in simpler regressions, but Active Share dominates after controls.
- Turnover is not a useful substitute for Active Share.

Why it matters. The paper's new measure is not just another concentration statistic. It is the most robust predictor among the active-management measures considered.

7 Short summary

- **New measure.** Active Share is the fraction of a fund's holdings that differs from its benchmark.
- **Main conceptual point.** Active management has two dimensions: stock selection and factor timing. Active Share captures the former; tracking error captures much of the latter.
- **Main data contribution.** The paper computes Active Share using fund holdings and holdings of nineteen realistic benchmark indexes from 1980 to 2003.
- **Main empirical trend.** The mutual fund industry became less active over time, with both explicit indexing and closet indexing rising in the 1990s.
- **Main performance result.** High Active Share funds outperform benchmarks; low Active Share nonindex funds underperform after fees.
- **Main investor lesson.** Do not pay active fees for a benchmark-like portfolio. If using active management, look for truly active funds, preferably smaller high Active Share funds with good recent performance.

- **Main research lesson.** Sorting all nonindex mutual funds together can hide strong heterogeneity. Distinguishing truly active funds from closet indexers changes the interpretation of mutual fund performance.

8 Conclusion

8.1 Core conclusion

Cremers and Petajisto show that Active Share is a simple but powerful measure of active management. It identifies whether a fund truly departs from its benchmark, and it predicts benchmark-adjusted performance in a way that tracking error does not.

8.2 Economic intuition

An active manager can beat a benchmark only by holding something different from that benchmark. Tracking error can be high because of broad factor tilts, but those tilts do not necessarily reveal stock-picking skill. Active Share directly measures the extent of benchmark deviation in holdings, so it better captures the possibility of value added from stock selection.

8.3 Takeaway

The paper's practical message is direct: Active Share separates truly active funds from closet indexers. The best evidence of skill appears among high Active Share funds, especially smaller funds and high Active Share funds with strong recent performance. The worst deal for investors is a fund that charges active fees while remaining close to its benchmark.